

Efficient PE Header Feature Extraction for Malware Analysis: Comparing Reduced and Extended Feature Sets

Fabiha Jalal¹, Sadia Tasnim Dhruba¹, Onamika Hossain¹ and Md Moniruzzaman¹

¹*Department of Computer Science and Engineering, Islamic University of Technology (IUT), Gazipur, Bangladesh
fabihajalal@iut-dhaka.edu*

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Abstract: Malware classification remains a critical challenge in cybersecurity as the volume and sophistication of malicious software continue to increase exponentially. Notably, prior studies on the APT1 dataset reported feature extraction times exceeding 19 minutes (1,157.81 seconds) for processing executable samples, making real-time analysis impractical. This research presents an efficient feature extraction methodology for static malware analysis utilizing Portable Executable (PE) header files. We conduct a comprehensive comparative study between two distinct feature set approaches: a reduced feature set (14 features) and an extended feature set (33 features) extracted from PE file headers, using the PE Characteristics field as the classification target. Our experiments, conducted on a newly created and publicly available dataset of 1,150 malware executable samples collected from MalwareBazaar, demonstrate that the extended feature set achieves the highest classification accuracy (94.35%) using Decision Tree, while Random Forest maintains 93.91% accuracy across both feature sets. Our proposed methodology reduces feature extraction time from approximately 19 minutes to approximately 215–281 seconds on our hardware, representing a substantial improvement over prior work.

1 INTRODUCTION

1.1 Overview

The proliferation of malware poses a critical threat to computer systems worldwide, with over 1 billion malware instances detected in 2023 alone, averaging approximately 33 new malware samples per second (Kalakuntla et al., 2019). The increasing frequency and sophistication of malware attacks necessitate the development of efficient and accurate detection methodologies. Malware analysis, the process of understanding the functionality and purpose of malicious software, plays a crucial role in identifying threats, assessing potential damage, and developing effective countermeasures.

Static malware analysis, which examines executable files without execution, offers several advantages over dynamic analysis, including safety, speed, and the ability to analyze malware that employs anti-analysis techniques (Black and Opacki, 2016). By leveraging structural information contained within Portable Executable (PE) files, static analysis can efficiently identify malicious characteristics without the computational overhead and security risks associated with runtime analysis.

1.2 Research Objectives and Contributions

This research aims to develop an efficient feature extraction methodology using PE header files that minimizes extraction time while maintaining classification accuracy. To this end, we conduct a comparative analysis between a reduced (14 features) and an extended (33 features) feature set, and evaluate six machine learning classifiers for malware classification based on PE header Characteristics. Additionally, we create and publish a novel malware dataset of 1,150 samples with pre-extracted PE header features for research purposes, and demonstrate the trade-offs between feature dimensionality, extraction time, and detection performance.

The primary contributions of this work are as follows. First, we propose a novel and efficient feature extraction technique that reduces extraction time from approximately 19 minutes to under 5 minutes. Second, we provide a publicly available dataset of 1,150 malware executable files with extracted PE header features. Third, we present a comprehensive comparative analysis of two feature set approaches across six machine learning classifiers, providing empirical evidence on the relationship between feature quantity

and classification performance.

2 BACKGROUND AND HISTORY

2.1 Malware Evolution

Malware, derived from “malicious software,” encompasses any program designed to damage, exploit, or gain unauthorized access to computer systems. The evolution of malware has progressed significantly since the emergence of early computer viruses in the 1980s. Modern malware categories include:

Viruses: Self-replicating programs that attach to legitimate files and propagate when infected files are executed.

Worms: Self-replicating malware that spreads through network connections without requiring host files.

Trojans: Malicious programs disguised as legitimate software that create backdoors, steal data, or enable unauthorized access.

Ransomware: Malware that encrypts victim files and demands payment for decryption keys. Notable examples include WannaCry (2017) and NotPetya (2017) (Choo and Dehghantanha, 2022).

Spyware: Software that covertly monitors user activities, captures keystrokes, and transmits sensitive information to unauthorized parties.

2.2 Malware Analysis Techniques

2.2.1 Static Analysis

Static analysis examines malware without execution by analyzing file metadata, extracting strings, and reading file headers (Santos and Festijo, 2021). This approach is safer, faster, and can identify malicious indicators that might be hidden during dynamic execution. Key components of static analysis include:

- File hash verification (MD5, SHA-256)
- PE header examination
- String extraction and analysis
- Disassembly and code pattern recognition

2.2.2 Dynamic Analysis

Dynamic analysis involves executing malware in controlled environments (sandboxes or virtual machines) to observe runtime behavior, including system call monitoring, network traffic analysis, file system modifications, and registry changes.

2.2.3 Hybrid Analysis

Hybrid analysis combines static and dynamic techniques to provide comprehensive malware characterization, leveraging the strengths of both approaches (Kumar et al., 2021).

2.3 Portable Executable (PE) File Format

The PE file format is the standard executable format for Windows operating systems, containing structural information essential for operating system loaders (Rezaei and Hamze, 2020). The PE structure begins with the DOS Header (MZ Header), which contains the magic number (0x5A4D) identifying the file as a DOS-compatible executable, followed by the DOS Stub that provides legacy DOS compatibility. The PE File Header contains critical metadata including machine architecture type, number of sections, timesamp, optional header size, and characteristics flags. The Optional Header, despite its name, is mandatory for executable files and contains entry point address, image base address, section and file alignment, operating system version requirements, subsystem type, DLL characteristics, and stack/heap size specifications (Maleki and Rastegari, 2019). Finally, the Section Table describes the layout of executable sections (.text, .data, .rdata, .rsrc, etc.). The PE header structure provides valuable features for malware classification, as malicious executables often exhibit distinctive header characteristics that differ from benign software (Al-Khshali et al., 2020; ?).

3 MOTIVATION

3.1 The Need for Efficient Feature Extraction

The exponential growth in malware variants demands analysis techniques that can process large volumes of samples efficiently. In 2021, ransomware attacks occurred approximately every 11 seconds, highlighting the critical need for rapid detection capabilities.

Previous research utilizing PE header features for malware detection, such as studies on the APT1 dataset (Balram et al., 2019), reported feature extraction times exceeding 19 minutes (1,157.81 seconds) for processing the entire dataset of executable samples. Such prolonged extraction times are impractical for real-time malware detection systems that must analyze thousands of samples daily.

3.2 Feature Selection Considerations

The selection of PE header features significantly impacts both extraction efficiency and classification accuracy. While comprehensive feature sets may capture more discriminative information, they also increase computational overhead. Conversely, minimal feature sets may sacrifice accuracy for speed.

This research investigates the optimal balance between feature comprehensiveness and extraction efficiency by comparing various numbers of features. We selected a reduced feature set of 14 attributes focusing on essential PE header fields that are most commonly referenced in prior malware detection studies, and an extended feature set of 33 attributes that incorporates all accessible `FILE_HEADER` and `OPTIONAL_HEADER` fields. These two sizes were chosen to represent a minimal, fast-extraction baseline versus a comprehensive, information-rich alternative. While other intermediate feature set sizes (e.g., 20 or 25 features) could also be explored, our goal was to establish the boundary conditions—the smallest practical set and the largest available set—to clearly quantify the trade-off between extraction time and classification accuracy.

Prior studies have demonstrated the effectiveness of PE header-based malware detection but have not adequately addressed the trade-off between feature quantity and extraction time, comparative analysis of different feature set sizes on the same dataset, or optimization of feature extraction for real-time detection applications. This research addresses these gaps through systematic experimentation and analysis.

4 WORKING MECHANISM

4.1 System Architecture

The proposed malware classification system comprises three primary phases: (1) Feature Extraction from PE files using the `pefile` library, (2) Machine Learning Classification using trained models, and (3) Classification Result output. Figure 1 illustrates this workflow.

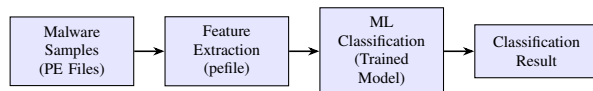


Figure 1: System Architecture Overview.

4.2 Dataset Preparation

4.2.1 Data Collection

Malware executable samples were collected from MalwareBazaar (MalwareBazaar, 2024), a reputable malware sample repository. The dataset comprises 1,150 PE executable files. The PE header `Characteristics` field, a bitfield encoding structural properties such as executable type and architecture, serves as the classification target. The dataset contains 30 distinct `Characteristics` values, forming a multi-class classification problem. The class distribution is imbalanced: the three most frequent classes account for approximately 60% of samples, while six classes contain only one or two samples each.

4.2.2 Data Processing

The Python `pefile` library was employed for PE header parsing. Files were validated to ensure proper PE format before feature extraction. Non-PE files (.rar, .elt, etc.) were excluded from the dataset.

4.3 Feature Extraction Methodology

4.3.1 Reduced Feature Set (14 Features)

The reduced feature set extracts 14 attributes from PE headers, with `Characteristics` used as the target variable. From the `FILE_HEADER`, two features are extracted: `Machine` and `NumberOfSections`. From the `OPTIONAL_HEADER`, twelve features are extracted: `AddressOfEntryPoint`, `ImageBase`, `SectionAlignment`, `FileAlignment`, `Subsystem`, `DllCharacteristics`, `MajorOperatingSystemVersion`, `MinorOperatingSystemVersion`, `MajorImageVersion`, `MinorImageVersion`, `SizeOfImage`, and `SizeOfHeaders`.

4.3.2 Extended Feature Set (32 Features)

The extended feature set extracts 33 attributes, with `Characteristics` used as the target variable. From the `FILE_HEADER`, six features are extracted: `Machine`, `NumberOfSections`, `TimeDateStamp`, `PointerToSymbolTable`, `NumberOfSymbols`, and `SizeOfOptionalHeader`. From the `OPTIONAL_HEADER`, twenty-seven features are extracted: `MajorLinkerVersion`, `MinorLinkerVersion`, `SizeOfCode`, `SizeOfInitializedData`, `SizeOfUninitializedData`, `AddressOfEntryPoint`, `BaseOfCode`, `ImageBase`, `SectionAlignment`, `FileAlignment`, `MajorOperatingSystemVersion`, `MinorOperatingSystemVersion`, `MajorImageVersion`, `MinorImageVersion`, `MajorSubsystemVersion`, `MinorSubsystemVersion`, `SizeOfImage`, `SizeOfHeaders`, `Checksum`, `Subsystem`, `DllCharacteristics`,

SizeOfStackReserve, SizeOfStackCommit, SizeOfHeapReserve, SizeOfHeapCommit, LoaderFlags, and NumberOfRvaAndSizes.

4.4 Feature Extraction Procedure

The feature extraction process is implemented using Python with the `pefile` library. Algorithm 1 presents the pseudocode for the extraction process.

Table 1: Algorithm 1: PE Header Feature Extraction.

Require: Directory path containing PE files
Ensure: CSV file with extracted features
1: for each file in directory do
2: Parse PE file using <code>pefile</code> library
3: if PE parsing successful then
4: Extract <code>FILE_HEADER</code> features
5: Extract <code>OPTIONAL_HEADER</code> features
6: Append features to dataset
7: else
8: Log error and skip file
9: end if
10: end for
11: Save dataset to CSV file

4.5 Machine Learning Classification

Six machine learning classifiers were employed for malware classification. Random Forest is an ensemble method using 100 decision trees with bootstrap aggregation. K-Nearest Neighbors (KNN) is an instance-based learning approach using Euclidean distance with $k = 5$. Decision Tree performs tree-based classification using information gain. Support Vector Machine (SVM) uses kernel-based classification with an RBF kernel. Logistic Regression performs linear classification with maximum likelihood estimation. Gradient Boosting is a sequential ensemble method with 50 estimators.

4.6 Experimental Setup

The dataset was split into 80% training (920 samples) and 20% testing (230 samples). Model validation was performed using 3-fold cross-validation, consistent across both feature sets. A random state of 42 was used for reproducibility. The evaluation metrics employed include Accuracy, Precision, Recall, and F1-Score.

5 RESULTS

5.1 Feature Extraction Performance

Table 2 presents the feature extraction performance comparison between the two approaches.

Table 2: Feature Extraction Performance Comparison.

Metric	Reduced (14)	Extended (33)
Extraction Time	215.81 sec	281.42 sec
Processing Speed	5.33 files/sec	4.09 files/sec
Files Processed	1,150	1,150
Errors	0	0
CSV Columns	16	35

Comparison with Prior Work (APT1 Dataset):

- APT1 PE Header Extraction: 1,157.81 seconds (~19 minutes)
- Our Extended Feature Set: 281.42 seconds (~4.7 minutes)
- **Improvement: 75% reduction in extraction time**

Note: This comparison is approximate, as the APT1 study used a different dataset, hardware environment, and extraction implementation. The figures illustrate the general magnitude of improvement achievable with our methodology rather than a controlled benchmark.

5.2 Classification Results - Reduced Feature Set

Table 3 presents the classification results for the reduced feature set (14 features).

Table 3: Classification Results - Reduced Feature Set (14 Features).

Classifier	Acc.	F1	Prec.	Rec.
Random Forest	93.91%	0.8379	0.8858	0.8264
Decision Tree	91.74%	0.7329	0.7681	0.7302
Gradient Boost.	90.87%	0.5956	0.6037	0.6028
KNN	75.22%	0.4636	0.4969	0.4590
Logistic Reg.	51.30%	0.1369	0.1309	0.1734
SVM	25.65%	0.0240	0.0151	0.0588

As shown in Table 3, tree-based classifiers dominate performance with the reduced feature set. Random Forest achieves the highest accuracy of 93.91% with an F1-score of 0.8379 and the best precision of 0.8858, indicating that ensemble methods effectively leverage even a limited set of PE header features for malware classification. Decision Tree follows closely

at 91.74% accuracy with an F1-score of 0.7329, while Gradient Boosting achieves 90.87% accuracy but with a notably lower F1-score of 0.5956, suggesting it struggles with certain malware classes despite reasonable overall accuracy. KNN shows moderate performance at 75.22%, indicating that distance-based methods have difficulty distinguishing Characteristics classes in the limited 14-dimensional feature space. Logistic Regression and SVM perform poorly at 51.30% and 25.65% respectively, which is likely attributable to the multi-class nature of the classification problem and the absence of feature scaling—both of which are critical for optimal performance of these linear classifiers. The notable gap between accuracy and F1-score across all classifiers (e.g., Gradient Boosting: 90.87% accuracy vs. 0.5956 F1) reflects the class imbalance in the dataset, where classifiers achieve high accuracy by correctly predicting the majority classes while struggling with underrepresented classes.

5.3 Classification Results - Extended Feature Set

Table 4 presents the classification results for the extended feature set (33 features).

Table 4: Classification Results - Extended Feature Set (32 Features).

Classifier	Acc.	F1	Prec.	Rec.
Decision Tree	94.35%	0.7724	0.7848	0.7651
Random Forest	93.91%	0.8377	0.9027	0.8086
Gradient Boost.	93.04%	0.7883	0.8420	0.7655
KNN	80.00%	0.5088	0.5304	0.5053
Logistic Reg.	41.30%	0.1098	0.1076	0.1368
SVM	25.65%	0.0240	0.0151	0.0588

With the extended feature set of 33 attributes, tree-based classifiers show notable improvement over the reduced set. Decision Tree achieves the best accuracy of 94.35% with the extended set, a 2.61 percentage point improvement. Random Forest achieves 93.91% accuracy with the highest F1-score of 0.8377 and precision of 0.9027, indicating strong performance despite matching its reduced-set accuracy—the additional features redistribute importance without improving overall accuracy for this ensemble method. Gradient Boosting reaches 93.04% with a substantially improved F1-score of 0.7883 compared to 0.5956 in the reduced set, indicating that the additional features help resolve previously ambiguous classifications. KNN benefits notably from the extended features, rising from 75.22% to 80.00%, a 4.78 percentage point improvement, which suggests

that the additional features create more meaningful distance separations in the higher-dimensional feature space. Notably, Logistic Regression drops from 51.30% to 41.30%, likely due to multicollinearity among the additional correlated features, which degrades the performance of linear models. SVM remains unchanged at 25.65%, confirming its unsuitability for this multi-class problem without proper feature scaling and kernel optimization.

5.4 Comparative Analysis

Table 5 presents the accuracy improvement achieved by the extended feature set across all six classifiers.

Table 5: Accuracy Comparison Between Feature Sets.

Classifier	Reduced	Extended	Improv. (pp)
Random Forest	93.91%	93.91%	0.00
Decision Tree	91.74%	94.35%	+2.61
Gradient Boost.	90.87%	93.04%	+2.17
KNN	75.22%	80.00%	+4.78
Logistic Reg.	51.30%	41.30%	−10.00
SVM	25.65%	25.65%	0.00

Table 5 reveals a clear pattern in how different classifier families respond to additional features. Decision Tree benefits the most among tree-based classifiers with a +2.61 percentage point improvement, achieving the highest overall accuracy of 94.35%. Random Forest shows no accuracy change (93.91% for both sets), suggesting that its ensemble mechanism already captures the discriminative information with just 14 features, though its F1-score and precision shift with the additional features. KNN improves by +4.78 percentage points, indicating that the additional features create more separable clusters in higher-dimensional space, enabling better distance-based classification. Gradient Boosting shows a +2.17 percentage point improvement, demonstrating that sequential ensemble methods also benefit from richer feature representations. In contrast, Logistic Regression suffers a 10 percentage point drop in accuracy with the extended set, highlighting the curse of dimensionality for linear models when features are correlated. SVM shows no change, remaining at 25.65% for both sets, which confirms that without proper feature scaling and hyperparameter optimization, kernel-based methods cannot effectively leverage PE header features for multi-class classification. These results suggest that the choice of feature set should be guided by the intended classifier—tree-based methods can work well with either set, while linear classifiers require careful feature engineering.

5.5 Feature Importance Analysis

Table 6 presents the top 10 most important features for malware classification using Random Forest on the extended feature set.

Table 6: Top 10 Feature Importance (Random Forest).

Rank	Feature	Importance
1	MajorLinkerVersion	0.088
2	TimeStamp	0.075
3	AddressOfEntryPoint	0.074
4	SizeOfCode	0.063
5	Machine	0.062
6	DllCharacteristics	0.059
7	SizeOfImage	0.058
8	ImageBase	0.054
9	SizeOfInitializedData	0.050
10	NumberOfSections	0.040

5.6 Performance Visualization - Reduced Feature Set

Figure 2 shows the classifier performance comparison for the reduced feature set.

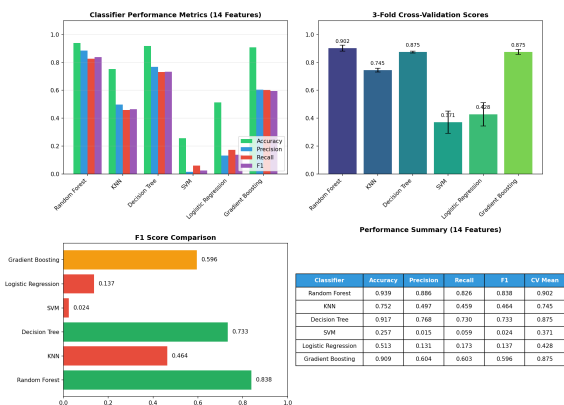


Figure 2: Classifier Performance Comparison - Reduced Feature Set (14 Features).

Figure 2 displays accuracy, precision, recall, and F1-score for all six classifiers using the reduced feature set. Random Forest dominates across all four metrics, while the sharp drop-off from Gradient Boosting to KNN illustrates the performance boundary between tree-based and non-tree-based methods on this feature set.

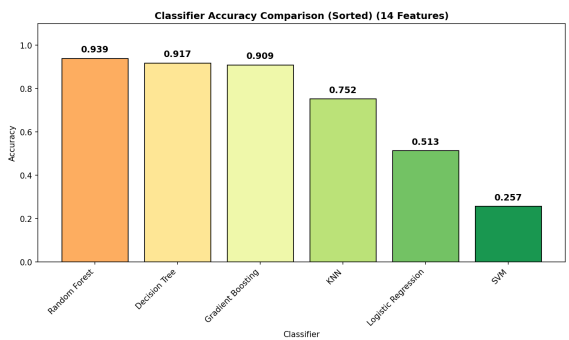


Figure 3: Accuracy Comparison - Reduced Feature Set (14 Features).

Figure 3 presents the accuracy values sorted in descending order. The top three classifiers (Random Forest at 93.91%, Decision Tree at 91.74%, and Gradient Boosting at 90.87%) are clearly separated from the lower-performing classifiers, with KNN at 75.22% forming a middle tier and Logistic Regression (51.30%) and SVM (25.65%) performing near or below random chance for this multi-class problem.

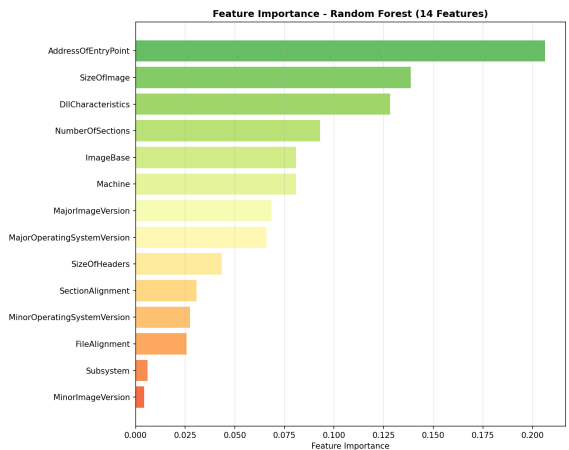


Figure 4: Feature Importance - Reduced Feature Set (14 Features).

Figure 4 shows the relative importance of each feature in the reduced set as determined by the Random Forest classifier. AddressOfEntryPoint emerges as the most important feature, followed by SizeOfImage and DllCharacteristics. The distribution indicates that a few features contribute disproportionately to classification decisions, suggesting potential for further feature reduction without major accuracy loss.

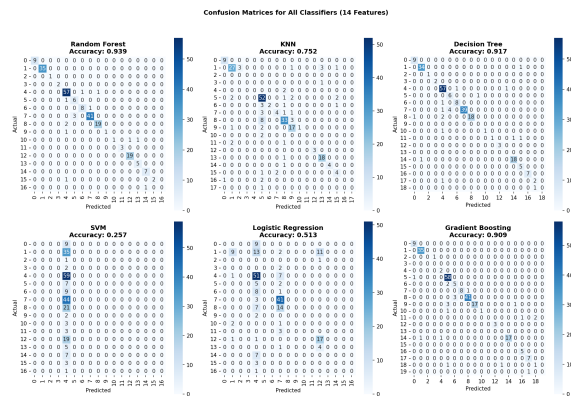


Figure 5: Confusion Matrices - Reduced Feature Set (14 Features).

Figure 5 presents the confusion matrices for all six classifiers. Random Forest and Decision Tree show strong diagonal dominance indicating correct classifications across most Characteristics classes, while SVM and Logistic Regression exhibit scattered predictions with many off-diagonal entries, confirming their inability to distinguish between classes effectively with this feature set.

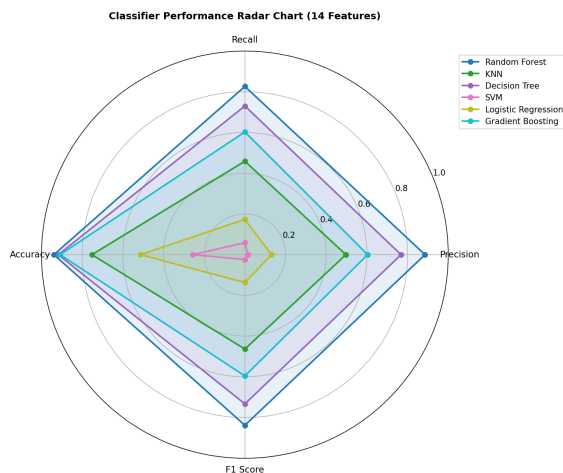


Figure 6: Performance Radar Chart - Reduced Feature Set (14 Features).

Figure 6 provides a radar chart visualizing precision, recall, accuracy, and F1-score simultaneously for all classifiers. Random Forest covers the largest area, indicating balanced and superior performance across all metrics. The collapsed shapes of SVM and Logistic Regression near the center confirm their poor multi-metric performance.

5.7 Performance Visualization - Extended Feature Set

Figure 7 shows the classifier performance comparison for the extended feature set.

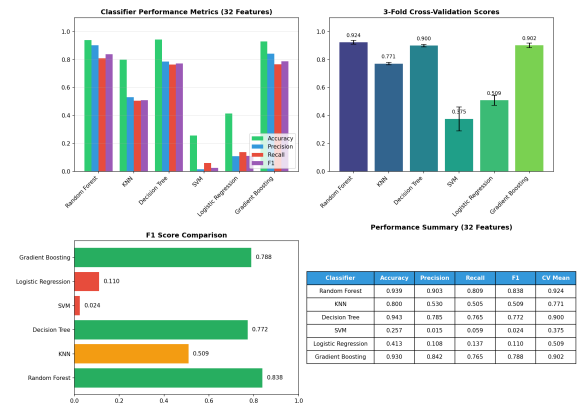


Figure 7: Classifier Performance Comparison - Extended Feature Set (32 Features).

Figure 7 displays the performance metrics for all classifiers using the extended 32-feature set. Compared to the reduced set, the bars for Random Forest, Decision Tree, Gradient Boosting, and KNN are visibly taller across all metrics, confirming the benefit of additional features for these classifiers. The gap between Gradient Boosting and KNN has narrowed, suggesting that the extended features help instance-based methods more proportionally.

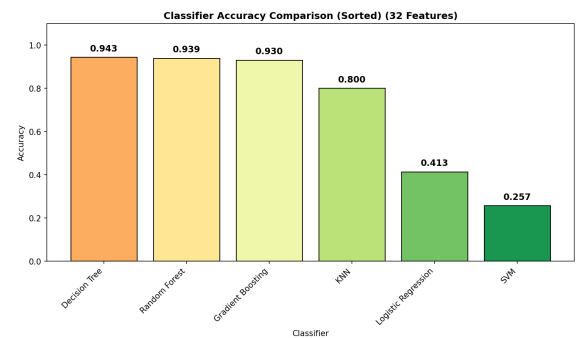


Figure 8: Accuracy Comparison - Extended Feature Set (32 Features).

Figure 8 shows the sorted accuracy comparison for the extended feature set. Decision Tree leads at 94.35%, followed closely by Random Forest (93.91%) and Gradient Boosting (93.04%). KNN's improvement to 80.00% is particularly notable. The persistent poor performance of Logistic Regression (41.30%) and SVM (25.65%) highlights that additional features do not compensate for fundamental

classifier limitations in this multi-class setting.

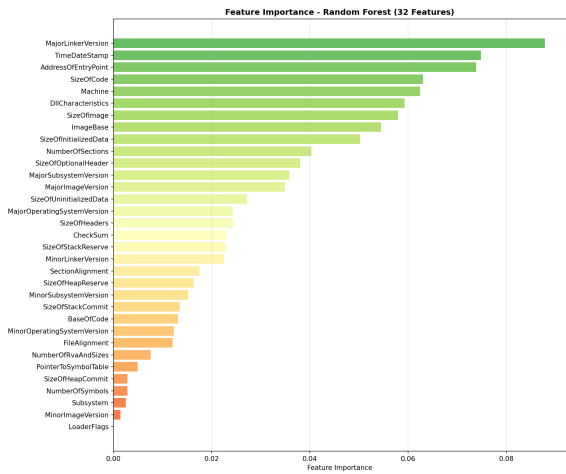


Figure 9: Feature Importance - Extended Feature Set (32 Features).

Figure 9 illustrates the importance ranking of all 33 features as determined by Random Forest. The distribution is more gradual compared to the reduced set, with MajorLinkerVersion, TimeDateStamp, and AddressOfEntryPoint leading. The long tail of features with small but non-zero importance values suggests that many of the additional features contribute incrementally to classification decisions, collectively accounting for the observed accuracy improvement.

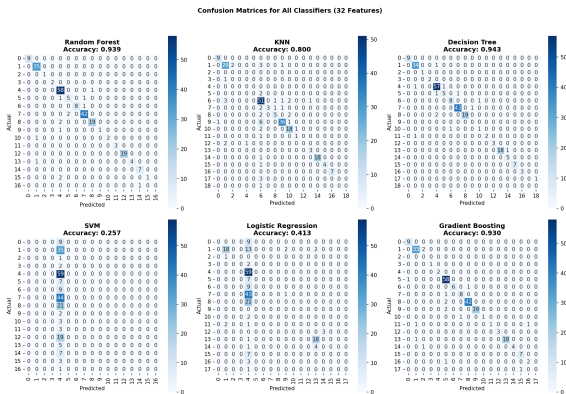


Figure 10: Confusion Matrices - Extended Feature Set (32 Features).

Figure 10 presents the confusion matrices for the extended feature set. Compared to Figure 5, the diagonal entries are more prominent for Random Forest, Decision Tree, and Gradient Boosting, indicating fewer misclassifications. KNN's confusion matrix also shows notable improvement with more concentrated diagonal predictions, while SVM and Logistic Regression remain largely unchanged.

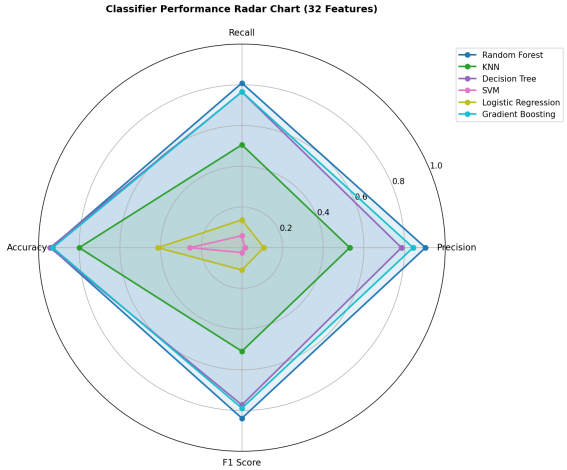


Figure 11: Performance Radar Chart - Extended Feature Set (32 Features).

Figure 11 shows the radar chart for the extended feature set. Random Forest's coverage area is the largest and most symmetric, reflecting balanced high performance. The expansion of KNN's shape compared to Figure 6 visually confirms its improvement with additional features, while SVM and Logistic Regression shapes remain near the center.

6 CONCLUSION

This research presents a comprehensive comparative study of PE header-based feature extraction for static malware analysis. The proposed methodology achieves a substantial reduction in feature extraction time, processing 1,150 malware samples in only 215–281 seconds depending on feature set size, compared to the 19-minute extraction time reported for the APT1 dataset in prior work. While this comparison spans different datasets and hardware environments, the magnitude of improvement suggests that our approach enables more practical deployment in real-time malware classification scenarios.

The extended feature set (33 features) outperforms the reduced feature set (14 features) for most classifiers. Decision Tree achieved the highest accuracy at 94.35% with the extended feature set, while Random Forest maintained 93.91% accuracy across both sets. KNN improved by 4.78 percentage points to 80.00%. However, Logistic Regression performed worse with the extended set (41.30% vs. 51.30%), likely due to multicollinearity among the additional features, and SVM exhibited poor performance overall due to the multi-class nature of the problem and the lack of feature scaling. While the extended feature set provides improved accuracy for most classifiers, it requires ap-

proximately 30% more extraction time (281 seconds vs. 215 seconds). For applications prioritizing speed over marginal accuracy improvements, the reduced feature set offers an effective compromise.

The research demonstrates that efficient malware classification based on PE structural properties is achievable using only PE header features without requiring computationally expensive dynamic analysis. The methodology is suitable for real-time malware scanning applications, large-scale malware dataset analysis, integration with existing antivirus systems, and automated malware triage systems.

Several limitations should be acknowledged. The classification target (Characteristics) is a PE header bitfield flag representing structural properties (e.g., executable type, architecture) rather than semantic malware family labels. Consequently, the classifier predicts structural categories of executables, not malware families. The dataset comprises 1,150 samples from a single source (MalwareBazaar), which may not fully represent the diversity of real-world malware and potentially introduces selection bias. The class distribution is imbalanced, with majority classes dominating accuracy while minority classes depress F1-scores. The dataset also lacks benign samples for binary classification evaluation. Static analysis cannot detect malware that employs runtime obfuscation or packing techniques, and PE header features alone may not capture behavioral characteristics of sophisticated malware. The methodology is specific to Windows PE executables and does not apply to other platforms. Additionally, SVM and Logistic Regression were not optimized with feature scaling or extensive hyperparameter tuning, and cross-validation was limited to 3 folds due to computational constraints.

7 FUTURE WORK

Future work should expand the dataset to include 10,000+ samples with benign files and labeled malware family annotations, explore feature selection techniques (PCA, t-SNE) to identify optimal feature subsets, implement parallel feature extraction using multi-core processors or distributed computing, develop ensemble models combining multiple classifiers for improved robustness, explore deep learning approaches (CNN, LSTM) for automatic feature learning (Kumar et al., 2021), and evaluate the methodology in a production-ready real-time malware classification system integrated with existing security infrastructure.

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